Can an LLM understand a game tileset?

Benchmarking LLM Tile Understanding and Tile Use • UCSC Augmented Design Lab • October 27, 2026

From recognizing tile relationships to generating coherent levels

Research proposal • Existing WFC/SAT experience • Invitation to collaborate

Three research questions

Framing proposed by Jim Whitehead

01 GENERATE

Given a tileset, can an LLM construct a coherent game level?

02 RELATE

Can it infer possible relationships among individual tiles and groups?

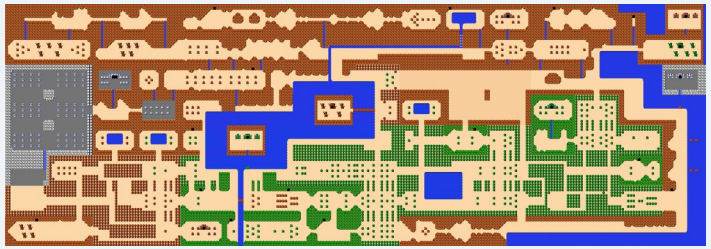
03 LEARN

How much does seeing example levels improve its performance?

Zelda: original and generated levels

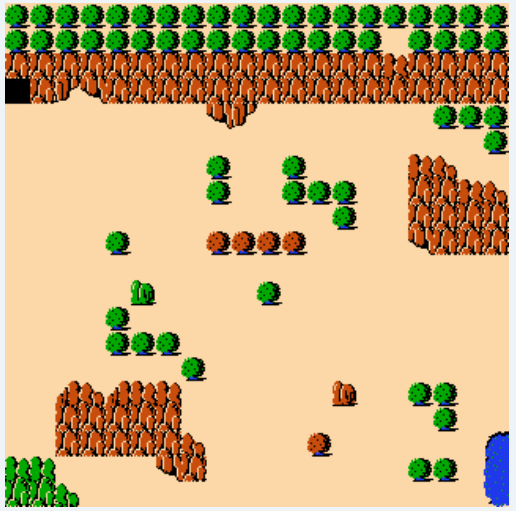
Existing WFC / SAT examples — motivation for testing higher-level tile understanding

Original Zelda map



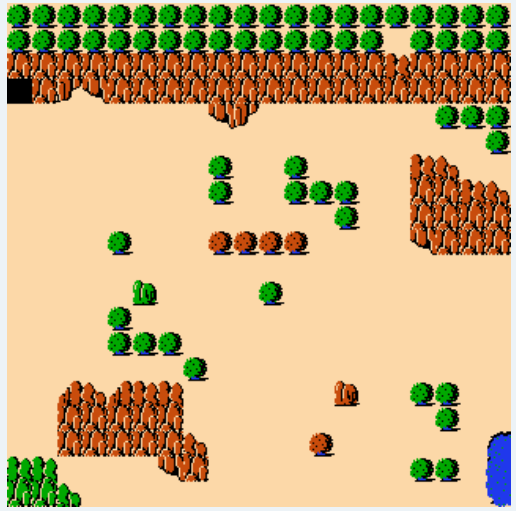
Reference example: complete map structures

Context-sensitive WFC



Generated output, seed 0

Context-sensitive WFC-as-SAT



Generated output, seed 0

Tiny Town: explicit house constraints also motivated the proposed relationship-inference tasks.

A tileset is not just a bag of images

Candidate relationships to ask a multimodal model to identify

ROLE

roof • wall • floor • door

LOCAL

which tiles connect at edges?

STRUCTURE

which groups form a house or room?

Predictions are hypotheses; evaluate them against annotated examples and explicit rules.

The central experiment: examples or no examples?

Hold model, task, tile IDs, and scoring constant

A — TILESET ONLY

Atlas + tile IDs + identical question or construction task.

B — TILESET + LEVELS

Same atlas and task, plus one or more example levels.

Initial candidates: Kenney Tiny Town, Tiny Dungeon, Tiny Farm; Zelda as another source.

Tasks and measurements

A small, reproducible benchmark before expanding coverage

MODEL TASKS

- Identify tile roles
- Predict adjacency or assemblies
- Generate a tile-ID grid
- Repair a broken arrangement

EVALUATION

- Valid IDs and grid dimensions
- Constraint satisfaction
- Precision / recall / F1 where labeled
- Blind human coherence rubric

Record prompts, inputs, model versions, outputs, and scoring for repeatable comparisons.

A small project that others can join

First milestone: one tileset, a few annotated tasks, and two input conditions

CONTRIBUTIONS

Choose examples and annotate relationships

Test prompts and models

Develop simple baselines

Review relevant papers

LATER POSSIBILITIES

Identify objects, shapes, and textures

Confirm inferred structure with a user

Pass structural rules to WFC or SAT

Collaboration welcome; scope and responsibilities remain open for discussion.

Resources and next steps

Proposed resources and selected reading — not a completed literature review

Tilesets: Kenney Tiny Town · Tiny Dungeon · Tiny Farm

Zelda map data: inventwithpython.com

Related work: Video Game Level Corpus · Tile Embedding

Patterns and PCG · GameTileNet · Learned Level Representations

github.com/stevex24/tile-understanding-benchmark

Next: select a first tileset, freeze tasks, annotate ground truth, and run baseline trials.